

Predicting Traffic Accident Severity in Brazil:
A Multiclass Machine Learning Approach with Imbalance
Handling

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Contents

1	Background and Introduction	3
1.1	The problem	3
1.2	Why this matters in Data Analytics	3
1.3	Problem definition & Project relevance	3
1.4	Research Questions	3
1.5	Hypothesis	4
1.6	Core technology and approach	4
2	Literature Review	4
2.1	Traffic Accident Severity Classification	4
2.2	Class Imbalance Methods	5
2.3	Multi-class vs Binary Classification	6
2.4	Explainability and Feature Importance	6
2.5	Tree-based Models on Tabular Data	7
2.6	Gaps and Contribution	7
3	Exploration of Data and Methods	8
3.1	Data Collection	8
3.2	Ethical Approval	8
3.3	Data Description	8
3.4	Data Cleaning	10
3.5	Feature Engineering	11
3.6	Exploratory Data Analysis	12
3.6.1	Target Variable Distribution	12
3.6.2	Geographic Patterns	13
3.6.3	Temporal Patterns	16
3.6.4	Accident Causes	16
3.7	Artefact Design	17
4	Proposed Future Process	18
4.1	Phase 1 – Finalising the dataset (mid/late June)	18
4.2	Phase 2 – Base modelling (late June)	18
4.3	Phase 3 – Imbalance-handling experiments (mid/late July)	19
4.4	Phase 4 – Hyperparameter tuning (early August)	19
4.5	Phase 5 – SHAP analysis (early/mid August)	19
4.6	Phase 6 – Writing and submission (late August–early September)	19
4.7	Final words	20
A	Project Links	21
B	Prompts for Generative AI	21

1 Background and Introduction

This section focuses mainly on introducing the problem, research questions, and the proposed solution that this work is going to perform.

1.1 The problem

In today’s world, vehicle transportation is almost indispensable. Nevertheless, it is one of the leading causes of death. In Brazil, it was confirmed to be the tenth most common cause of death, which annually has confirmed to have around 33,000 deaths (Alves et al. 2021). The cost of traffic accidents in Brazil is around 220 billion reais per year, which accounts for up to 4% of Brazil’s GDP (Nunes & Silva 2021). Furthermore, studies are mainly focused on well-developed countries. While leaving the low- and middle-income countries behind, despite being one of the biggest causes of traffic accidents (Alves et al. 2021). This is a problem which needs to be addressed. Otherwise, it could lead to much more serious consequences.

1.2 Why this matters in Data Analytics

Accident severity prediction is a broad and well-studied field, which is ideal for the first and proper machine learning dissertation. The Brazilian PRF accident dataset contains around 500,000 data points, which should be capable of performing meaningful prediction models. Practical results could also contribute to better resource allocation and achieving a higher success rate in saving lives.

1.3 Problem definition & Project relevance

Most of the previous studies were focusing on binary classification and acknowledged imbalance data or only performed one technique (Scholz et al. 2024, Gabardo et al. 2026). This work would like to take a different approach. It would like to handle three severity classes: With dead victims, With injured victims, Without victims and apply several techniques to handle the imbalanced data to achieve practical results.

1.4 Research Questions

This study wants to address three research questions:

1. RQ1: When it comes to predicting the multiclass severity of accidents using macro-F1 and per-class metrics, can tree-based classifiers (Decision Tree, Random Forest, XGBoost, LightGBM, and CatBoost) outperform a Logistic Regression baseline?
2. RQ2: How does handling class imbalance (using random oversampling, random undersampling, SMOTE, ADASYN, SMOTE+Tomek, and class weighting) affect the performance of these models, and how do the main methods of class imbalance handling compare?
3. RQ3: Which features are most important in predicting traffic accident severity?

1.5 Hypothesis

The imbalance handling methods will improve models and their results over the base non-handling models. SHAP is going to select and identify the dominant predictors which are the biggest contributors to traffic accidents.

1.6 Core technology and approach

This work is going to be recorded by version control software such as GitHub, where all commits are going to be well commented. It will use Visual Studio Code, which is an Integrated Development Environment, and their Python libraries such as scikit-learn, imbalanced-learn, XGBoost, LightGBM, CatBoost, and SHAP. Lastly, the six models are going to be used, and the best three of them are going to be compared with seven imbalance-handling techniques. The final paper will be written on the website called Overleaf. AI will be used for explanation, necessary or time-consuming tasks. Grammarly, Google Translate, and DeepL will be used for writing, spell-checking, and looking up words.

2 Literature Review

This section outlines the rationale and justification for the methods that will be used in the upcoming study.

2.1 Traffic Accident Severity Classification

When it comes to predicting the severity of an accident, it is important to consider which methods to use. Statistical models, like logit, probit, ordered probit, and multinomial logit, were used before as the most conventional approach to crash-severity analysis, but recent research papers and tests discovered that machine learning models can outperform them on predictive accuracy ([Zhang et al. 2018](#)). Machine learning models are able to handle non-linear interactions and high-dimensional categorical features, which makes them more ideal for predictions than statistical models.

Machine learning models predicting accident severity can be split into two categories. Supervised (learns from labels) or unsupervised (finds patterns without labels), where the standard approach is to use supervised for this kind of topic. Deep learning models are commonly used on image data, rather than tabular. Therefore, the classical supervised machine learning models are the main topic to be used on severity accidents data.

[Zhang et al. \(2018\)](#) primarily compared two statistical models for crash severity prediction with four machine learning models. Although all machine learning models achieved better results, they suffered from overfitting. This did not occur with the statistical models, but their results for minority prediction was 0%, which is insufficient. [Zhang et al. \(2018\)](#) concluded that higher accuracy doesn't mean trustworthy insights, because machine learning models can disagree with each other on variable importance and conclusion. However, this can provide further discussion.

Other studies [Lee et al. \(2021\)](#) predicted the severity of crashes and identify possible risk factors (human, vehicle, road, environmental, etc.) while using only a single CART decision

tree. Data from the years 2004 - 2017 for around 170,000 US city crashes were used to predict the final four severity levels. While achieving a prominent 65.78% accuracy, the overall result led to a finding that the model failed to predict any fatal injuries (0% precision and 0% recall).

Labib et al. (2019) were able to achieve 80% accuracy on four-class severity. However, the dataset consists primarily of data related to fatal accidents, which means that the main results will pertain more to the prediction of the majority class than to performance on the minority class. This is a class imbalance issue, which can lead to bias when implemented.

Two recent studies targeted the Brazilian PRF dataset and applied machine learning models to predict accident severity. Scholz et al. (2024) performed binary predictions on five classifiers, such as Decision Tree, Random Forest, KNN, Naive Bayes, and MLP, on data from the year 2017 to 2022 on the Brazilian PRF dataset, enhanced by including vehicle prices as well, while stating the limitation of not performing any imbalance technique, which is a problem for prediction of minority classes. Meanwhile, Gabardo et al. (2026) applied SMOTE (Synthetic Minority Over-sampling Technique), which creates synthetic samples by interpolation between existing data points rather than duplicating minority-class instances. This was used for three classifiers (RF, KNN, MLP), but its limitation lies in the fact that it focused on only one region (southern Brazil) and was used only for binary classification.

Outside the Brazilian context, Ahmed et al. (2023) demonstrated that meaningful metrics are possible to achieve while focusing on using techniques such as SMOTE oversampling, ensemble methods and SHAP for explainability, which leads to a reliable and trustworthy implementation of the results.

These findings can be considered as a methodological stepping stone for this study, which aims to fill the gaps of previous studies by predicting severity across multiple classes using SMOTE oversampling technique, focusing on the entire country rather than just a specific region.

2.2 Class Imbalance Methods

Studies to this point confirm that, when it comes to predicting performance, it is necessary to use appropriate techniques to address class imbalance. Disadvantages of imbalance data can lead to meaningless accuracy, models predict majority classes, per-class metrics collapse on the minority class. The approaches to handle imbalanced data are (Narwane & Sawarkar 2019):

- Data-level - modify the training set (over/under-sampling)
- Algorithm-level - modify the classifier itself
- Cost-sensitive - penalise minority-class errors more heavily
- Ensemble-based - combine multiple weak learners with imbalance-aware schemes

One of the foundational data-level methods is SMOTE, which helps the model develop a better general understanding of decisions, rather than simply memorising recurring events. SMOTE is a technique to use on imbalanced data for machine learning models to identify fraud detection, medical diagnosis, and crash severity prediction (Chawla et al. 2002).

Although SMOTE is an important technique, it treats all minority classes equally, even though some of them are easier for a model to learn. However, ADASYN (Adaptive Synthetic) is able to generate synthetic samples for minority classes like SMOTE, but targets specific areas, which are harder to learn, thereby improving the model’s decision boundary. A limitation of using ADASYN is reduced precision, but in return, recall is increased (He et al. 2008).

To handle noise and class overlap, the hybrid methods such as SMOTE- Tomek and SMOTE-ENN exist. This cleans up ambiguous or noisy samples and enhances performance (Narwane & Sawarkar 2019).

Tanha et al. (2020) performed a benchmark on 19 multi-class datasets to compare methods on imbalanced data. It was discovered that SMOTEBoost consistently outperformed RUSBoost across the benchmark and native multi-class boosters, such as CatBoost and LightGBM, outperformed binary-decomposition OVO approaches.

This subsection of literature points out the gaps in other researchers’ papers for crash-severity studies, where most of them do not handle the imbalanced data and are focused on binary classification rather than on multiple classifications, which is under research. This also confirms the methodological approach of this dissertation, which favours the SMOTE method over the RUS method, compares different kinds of imbalanced handling techniques and highlights the arguments for choosing multiclass classification over binary classification, which is further elaborated in the following section.

2.3 Multi-class vs Binary Classification

Previous studies on the severity of traffic accidents have focused on the use of a binary severity classification (Scholz et al. 2024, Gabardo et al. 2026). Although predicting accurate results is essential, binary classification loses important information that could aid in emergency response, resource planning, and policy-making, enabling an appropriate response based on multiple classes, which is more suitable for real-life world scenarios. The difficulty with multi-classification problems comes with handling imbalanced data. Minority classes are more difficult to handle, multi-class imbalance receives less attention from the model and receives less research attention than binary classification (Narwane & Sawarkar 2019). Despite the increased difficulty of performing a multi-class classification model, there is evidence that native multi-class boosting outperforms binary-decomposition (OvO/OvR) approaches on imbalanced data (Tanha et al. 2020). There is a visible natural order in terms of severity. The previous empirical work discovered that using a nominal multi-class model outperformed ordinal models, which leads to the conclusion of using nominal models, even though the data are ordinal (Zhang et al. 2018). Based on these findings, a methodological approach is being developed that focuses on the three-category nominal problem, with the challenge of dealing with data imbalance being addressed.

2.4 Explainability and Feature Importance

Simple explainability is important to offer for stakeholders in order to clearly define trust, accountability, verifiability, and ethical standards. It is necessary to be able to understand the model prediction. Models with higher accuracy and black-box systems are less valuable than

explainable models with less accuracy. The insight is a key element to understand. Without it, unknown errors can be created and can undermine the credibility of the parties involved. Therefore, to be able to assign value to each feature for each individual prediction, SHAP (SHapley Additive exPlanations) is going to be used, which is a method to explain the output of any machine learning model. LIME (Local Interpretable Model-Agnostic Explanations), which is also a method to explain the output of any machine learning model and raw feature importance do not properly guarantee satisfaction of each desirable property, such as local accuracy, missingness, and consistency (Lundberg & Lee 2017). The LIME method allows only local interpretability, whereas the SHAP method can provide both local explanations and global feature rankings, which is more beneficial for answering research question 3. Ahmed et al. (2023) and Scholz et al. (2024) used SHAP to find the most appropriate predictor in their studies, which also supports the decision of using it on tree-based models.

Therefore, this leads to the conclusion to use SHAP for identifying the important predictors for accident severity data, revealing non-obvious patterns and providing proper insight for stakeholders or interested parties.

2.5 Tree-based Models on Tabular Data

Deep learning or neural network models are becoming a standard approach in many fields. Shwartz-Ziv & Armon (2022) was capable of proving that machine learning algorithms for structured data, such as XGBoost, are capable of dominating 8 out of 11 datasets against four "deep tabular" models (TabNet, NODE, DNF-Net, 1D-CNN). Each deep learning model was capable of performing well on its individual paper, but was unable to adapt to other possibilities. Grinsztajn et al. (2022) findings led to the conclusion that tree-based models are the most appropriate to use on medium-sized datasets based on 45 benchmark datasets testing. This led to the decision to use Random Forest, XGBoost, LightGBM, and CatBoost as the main models with logistic regression as the baseline model for performance evaluation, thereby achieving faster results.

2.6 Gaps and Contribution

This literature review addresses three gaps that this dissertation intends to fill.

- Multi-class on Brazilian PRF data

Previous empirical studies have focused on identifying binary classifications. However, they did not address severity as a problem for multidimensional classification, but merely mentioned it as a topic for future research (Scholz et al. 2024, Gabardo et al. 2026).

- Systematic imbalance-method comparison

Prior work tends to ignore, acknowledge or only apply one method of handling imbalance data (Ahmed et al. 2023, Gabardo et al. 2026, Lee et al. 2021, Labib et al. 2019). It lacks comparison of using multiple methods across three classifiers, which this dissertation intends to fill (Tanha et al. 2020, Narwane & Sawarkar 2019).

- Modern explainability on PRF data

This paper focuses on the TreeSHAP algorithm, a specific variant of the SHAP algorithm that is capable of achieving the best performance for multi-class models on the PRF national dataset, which other studies did not elaborate on or investigate ([Scholz et al. 2024](#), [Gabardo et al. 2026](#)).

By answering the research questions, this paper addresses the gaps that the literature review established.

3 Exploration of Data and Methods

The preprocessing method and EDA findings will be described and discussed below.

3.1 Data Collection

The data were collected by the Brazilian Federal Highway Police (PRF) ([LippoM 2024](#)). They are publicly available on the Kaggle repository. This study will cover a period of seven years of data collection. More specifically, from 2017 to 2023, which approximately contains 463,152 records. All the information about the traffic accidents is processed annually, filled in CSV format and merged together. This process occurred in each state, meaning that none were excluded. The data is geographically imbalanced because it is collected only when a traffic accident occurs. This should be taken into account, as future plans do not include handling the issue of geographical imbalance.

3.2 Ethical Approval

The data were collected by the Brazilian government and are publicly available. The dataset on Kaggle is fully anonymised and does not present any personally identifiable information. These factors prevent unethical behaviour and lead to the conclusion that no user approval is required. This work acknowledges the use of Brazilian open government data for machine learning purposes, academic reuse, analysis, and publication.

3.3 Data Description

As mentioned in the literature review, this dataset is considered medium-sized. It consists of 463,152 rows and contains 27 columns, which were collected from 01/01/2017 to 20/08/2023.

Table 1: Description of the 27 raw columns in the PRF dataset.

Column	Type	Description
inverse_data	str	Date of the accident (originally in reversed format)
week_day	str	Day of the week (Monday–Sunday)
hour	str	Hour at which the accident occurred
weather_timestamp	str	Phase of the day (sunrise, day, sunset, night)
state	str	Brazilian state of occurrence (e.g. SP, RJ, MA)
road_id	float64	Federal highway identifier (e.g. 116, 101)
km	str	Kilometre marker along the highway
city	str	City where the accident occurred
road_direction	str	Direction relative to the kilometre marker (increasing / decreasing)
latitude	float64	GPS latitude of the accident
longitude	float64	GPS longitude of the accident
regional	str	PRF regional administrative district
police_station	str	PRF police station that recorded the accident
cause_of_accident	str	Reported cause of the accident (80+ raw categories)
type_of_accident	str	Collision type
weather_condition	str	Weather at time of accident
road_type	str	Road layout type (single lane, double lane, etc.)
road_delineation	str	Lane marking and delineation condition
people	int64	Total number of people involved
deaths	int64	Number of fatalities
slightly_injured	int64	Number of people with minor injuries
severely_injured	int64	Number of people with serious injuries
uninjured	int64	Number of people unhurt
ignored	int64	Number of people whose condition was not reported
total_injured	int64	Sum of slight and severe injuries
vehicles_involved	int64	Number of vehicles involved
victims_condition	str	Severity classification: Without / With injured / With dead victims

The columns can be labelled and categorised for better clarification and description. This will result in 5 categories.

Table 2: Raw columns of the PRF dataset grouped by category.

Category	Count	Columns
Temporal	4	<code>inverse_data</code> , <code>week_day</code> , <code>hour</code> , <code>weather_timestamp</code>
Location	9	<code>state</code> , <code>road_id</code> , <code>km</code> , <code>city</code> , <code>road_direction</code> , <code>latitude</code> , <code>longitude</code> , <code>regional</code> , <code>police_station</code>
Accident Circumstance	5	<code>cause_of_accident</code> , <code>type_of_accident</code> , <code>wheather_condition</code> , <code>road_type</code> , <code>road_delineation</code>
Outcome	8	<code>people</code> , <code>deaths</code> , <code>slightly_injured</code> , <code>severely_injured</code> , <code>uninjured</code> , <code>ignored</code> , <code>total_injured</code> , <code>vehicles_involved</code>
Target	1	<code>victims_condition</code>
Total	27	

The target variable for this project is *victims_condition*, which is a three-class categorical variable (Without victims, With injured victims, With dead victims). The category With injured victims has the largest share of this target variable, representing approximately 71% of the total data. In contrast, the smallest category is “With dead victims”, which represents approximately 6.8% of the total data. This should provide a general understanding of how imbalanced the target variable is. The attributes in category *Outcome* are unusable because they contain variables after the accident happened, which is inadequate for model prediction.

3.4 Data Cleaning

It is important to check the data’s credibility. Data cleaning is the first step toward verifying the reliability of the data and, if necessary, improving it. As seen in Table 1, it is apparent that the column labels are insufficient. For example, `inverse_data`, `weather_timestamp`, and `wheather_condition` contain grammatical errors and ambiguities. Therefore, the step was taken to rename all columns for greater clarity and readability. The result is visible to see in Table 3.

During the examination of data. It was decided to combine the Date and Hour column. A single column, Datetime, was created, and other columns were dropped. This can prevent redundancy and should be easier to use. This column was also moved to the front of the dataset for better readability. The column Road_Km has a data type string, which is inappropriate. Machine learning models are not capable of processing text. Therefore, the type was changed to be numeric.

The following step was to handle missing values. It was discovered that columns with missing values are Road_ID, Road_Km, Regional, Police_Station. The column with the smallest amount of missing values is Regional, which contains 10 of them. It was decided to drop those rows, because the amount would have no effect on future model performance. Assuming that the Road_ID and Road_Km variables are connected, which was based on the number of missing values (990), the step to nullify this issue was to fill the missing values with NO_ROAD for Road_ID and -1 for Road_Km. Furthermore, it was decided to create a column (Missing_Road_Data) which will directly show the model that these sections contain missing values. The last column with missing values was Police_Station. The same approach was taken. The missing values were filled with UNKNOWN, and the Missing_Police_Station column was cre-

ated for the model to identify the missing values.

One of the last steps to complete the Data cleaning was to check for duplicates. 9 duplicates were found, and based on their content, it was decided to remove them. None of them showed anything significant or unusual that would have led to the decision to keep them.

The last step was to check the outliers. The result is visible to see in 4. Only Latitude and Longitude outliers were dropped. The invalid records are not necessary to keep. The other outliers serve as examples of rare but valid results that represent unusual cases. Dropping them would be insufficient for the final model's outcome. It should be noted that columns with outliers will most likely be excluded, as they are not suitable for model prediction.

This way, the data cleaning was completed, and the work can continue for the next phase.

Table 3: Column renaming applied to the raw PRF dataset.

Original	Renamed	Reason
inverse_data	Date	Misleading name, standardised
week_day	Day	Simplified
hour	Hour	Casing
state	State	Casing
road_id	Road_ID	Casing
km	Road_Km	Clarified meaning
city	City	Casing
cause_of_accident	Accident_Cause	Reordered for consistency
type_of_accident	Accident_Type	Reordered for consistency
victims_condition	Victims_Condition	Casing
weather_timestamp	Day_Phase	Misleading: held day phase, not a timestamp
road_direction	Km_Direction	Clarified meaning
wheather_condition	Weather_Condition	Corrected spelling typo
road_type	Road_Type	Casing
road_delineation	Road_Delineation	Casing
people	People	Casing
deaths	Deaths	Casing
slightly_injured	Slightly_Injured	Casing
severely_injured	Severely_Injured	Casing
uninjured	Uninjured	Casing
ignored	Ignored	Casing
total_injured	Total_Injured	Casing
vehicles_involved	Vehicles_Involved	Casing
latitude	Latitude	Casing
longitude	Longitude	Casing
regional	Regional	Casing
police_station	Police_Station	Casing

3.5 Feature Engineering

During Exploratory Data Analysis (EDA), 11 feature columns were created for a proper understanding of temporal, geographic, and categorical structure, which were not available to use in the original data.

Table 4: IQR-based outlier detection across numeric columns.

Column	Outliers	Share
People	8,586	1.9%
Deaths	31,344	6.8%
Slightly_Injured	20,283	4.4%
Severely_Injured	96,750	20.9%
Uninjured	37,195	8.0%
Ignored	57,240	12.4%
Total_Injured	30,484	6.6%
Vehicles_Involved	8,573	1.9%
Road_Km	4,356	0.9%
Invalid coordinates (dropped)	50	0.01%

Temporal features were created with the help of a column called Datetime. These features help to analyse the time pattern of accidents. One feature that needs to be explained is Day_Type. One of the topics considered for exploration during EDA was whether or not Saturdays and Sundays are equally deadly. The results led to the conclusion that Sundays are deadlier than Saturdays, and therefore, the Day_Type type was created to address this issue.

The Accident_Cause column has over 80 categories, which are not equally split. This was solved by merging certain categories to create a column with only 11 broader categories. This creates a low cardinality categorical predictor, which should be more useful.

Similarly, the Region_BR was created with five official macro regions from the State column. States were assigned to official regions based on the Brazilian Institute of Geography and Statistics (IBGE). This helped reveal that the North and Northeast are deadlier than other regions.

Lastly, as mentioned before, the Missing_Road_Data and Missing_Police_Station columns were created to help the model identify missing values. Possibly, the missing data are not random and could give important information about reporting quality and accident context.

The overall results are visible in Table 5. This table summarises the features described above.

3.6 Exploratory Data Analysis

The EDA is trying to explore the clean dataset while inspecting and interpreting the meaning of the plots and results. After cleaning, the dataset contains 463,133 records.

3.6.1 Target Variable Distribution

The target variable exhibits a highly imbalanced distribution. As mentioned previously, the minority class is the one with dead victims, which contains around 6.8% of the overall data, and the most common class is the one with injured victims, which contains 71.7%. If a naive classifier were used, the one that would ignore the features input, it would predict accuracy to be 71.7%, which is correct by the total share of accidents, but it would lack proper understanding. Therefore, this is the main motive for the second research question. It also illustrates that accuracy is not the proper metric to use in this case. Therefore, the per-class metrics such as F1, Macro-F1 and Matthews Correlation Coefficient (MCC) will be used to compare the

Table 5: Engineered features derived from the raw PRF dataset.

Feature	Derived from	Type	Purpose
Datetime	Date + Hour	timestamp	Unifies date and hour into a single field
Hour	Datetime	integer (0–23)	Captures hour of day pattern
Month	Datetime	integer (1–12)	Captures seasonality
Weekday	Datetime	categorical (7)	Day name, replaces the raw Day column
Is_Weekend	Datetime	boolean	Quick binary weekend indicator
Day_Type	Datetime	categorical (3)	Weekday / Saturday / Sunday splits the weekend because Sat and Sun differ
Time_Period	Hour	categorical (4)	Morning / Afternoon / Evening / Night
Cause_Group	Accident_Cause	categorical (11)	Collapses 80+ causes into broader categories
Region_BR	State	categorical (5)	Maps 27 states to 5 IBGE macro regions
Missing_Road_Data	Road_ID	boolean	Preserves missingness signal for Road_ID
Missing_Police_Station	Police_Station	boolean	Preserves missingness signal for Police_Station

results as well. [Tanha et al. \(2020\)](#) mentioned that MCC is the best metric to use on multiclass imbalanced data.

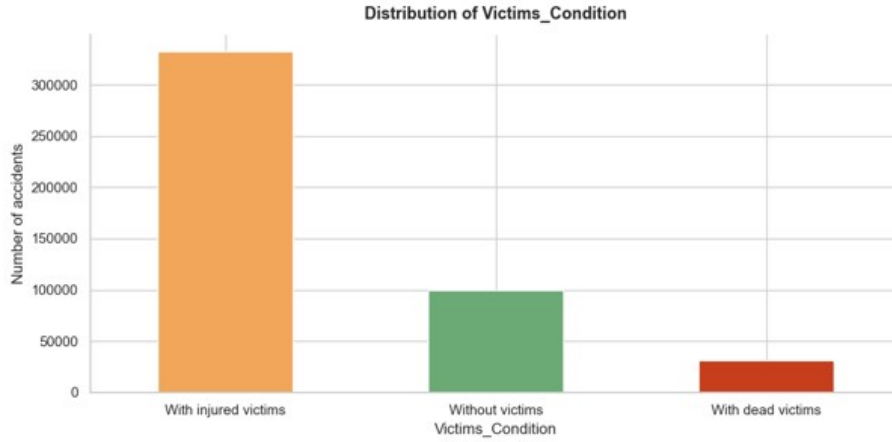


Figure 1: Distribution of the target variable `Victims_Condition`

3.6.2 Geographic Patterns

During the data examination, there was a step to aggregate states into broader categories. Macro-regions were created to find visible patterns, which would be invisible with 27 states. When each record was mapped based on the 5 IBGE macro-regions, it was discovered that Accidents in the north east tend to exhibit double the fatality rate as those in the south. This result leads to the assumption that most likely the `Region_BR` column will be an important predictor. This connects to research question 3, where SHAP will be used for the analysis. [Scholz et al. \(2024\)](#) found that the region category is one of the most important predictors for the PRF dataset. This should confirm that the hypothesis is correct and that it will prove to

be true.

Table 6: Fatality and injury profile across the five Brazilian macro-regions.

Region_BR	Accidents	Injured share	Fatal share
Northeast	98,076	68.4%	10.2%
North	26,182	70.1%	8.9%
Centre-West	56,516	69.6%	7.0%
Southeast	141,603	74.8%	5.7%
South	140,756	72.1%	5.1%

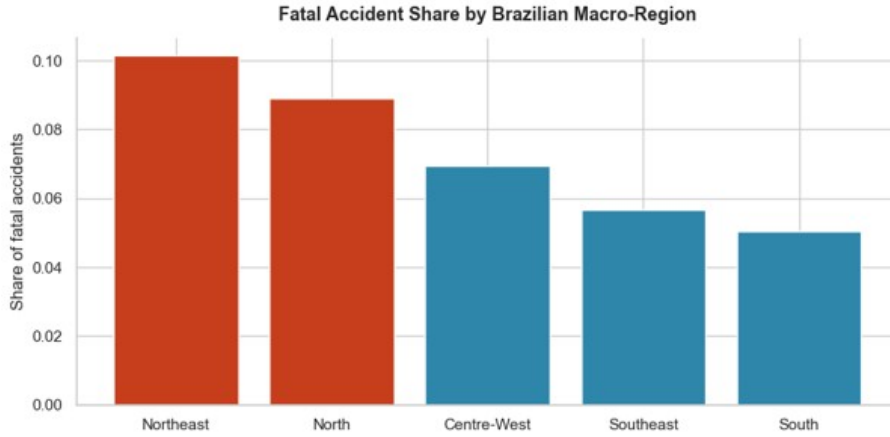


Figure 2: Fatal share by Brazilian macro-region.

An attribute that is potentially more specific than the region or state itself is Road_ID. The map created in Figure 3 shows where fatal accidents occur most frequently. A total of 7,917 fatal accidents occurred on highways BR-116 and BR-101. It is important to mention that these roads are among the longest in Brazil. It is understandable why their numbers are this high. [Ahmed et al. \(2023\)](#) says that road is one of the most important predictors of severity for New Zealand crash data. This gives an opportunity to assume the same applies to this dataset. Therefore, the feature is expected to be highly valuable for the SHAP analysis.

Table 7: Top ten Brazilian federal highways by fatal accident count

Federal highway	Accidents with dead victims
BR-116	4,097
BR-101	3,820
BR-153	1,394
BR-381	1,140
BR-040	1,099
BR-316	1,087
BR-364	1,029
BR-163	982
BR-230	842
BR-277	838

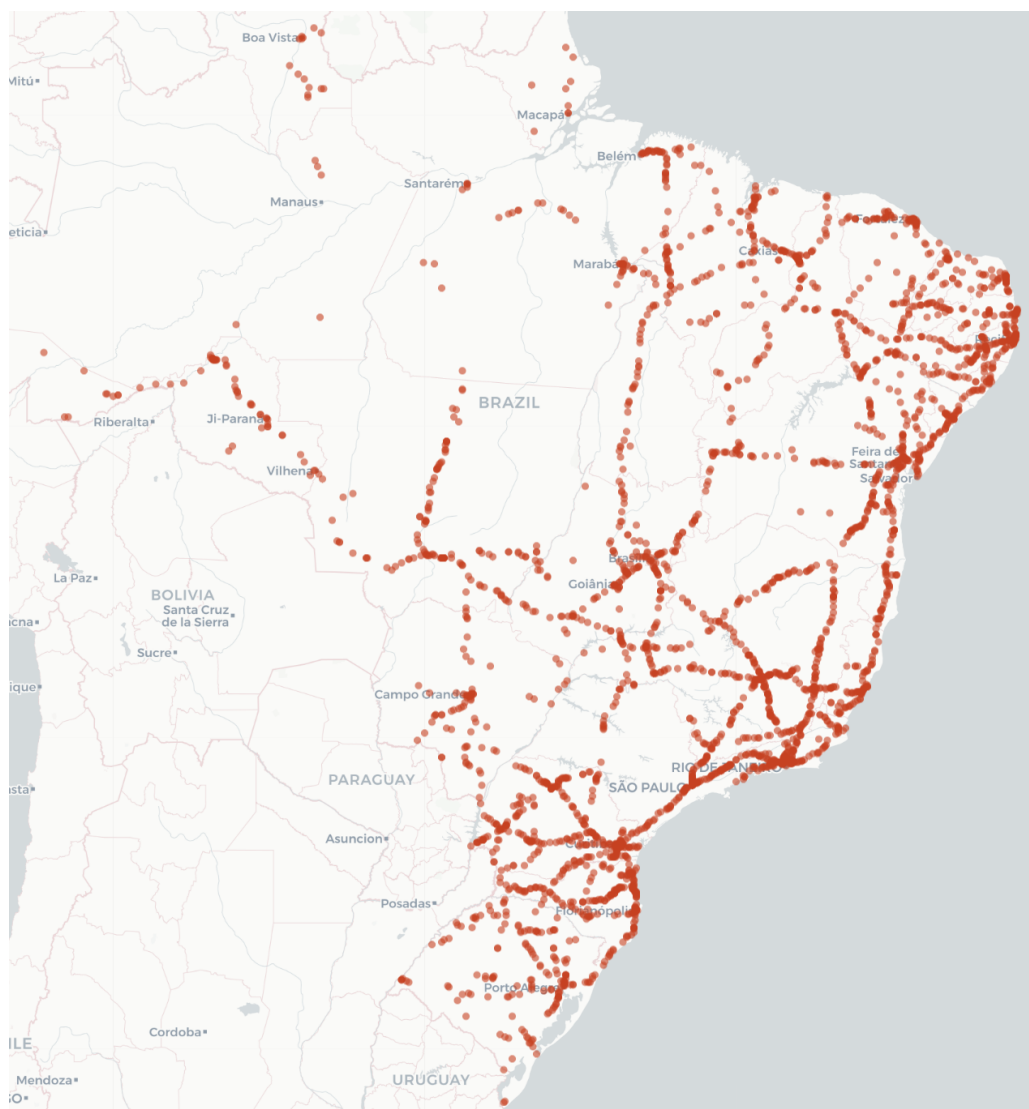


Figure 3: Spatial distribution of fatal accidents on Brazilian roads.

3.6.3 Temporal Patterns

The EDA of this theme found that there are different results when comparing weekdays and weekends, even though the days are evenly distributed. Furthermore, it was discovered that Weekends are more deadly than weekdays, especially during the night hours (22:00–05:00). What was also noticeable was that the results for Saturdays and Sundays differ, with Sundays being more deadly. This should be an adequate justification for retaining the Day_Type attribute, which has three categories (Weekday, Saturday, Sunday).

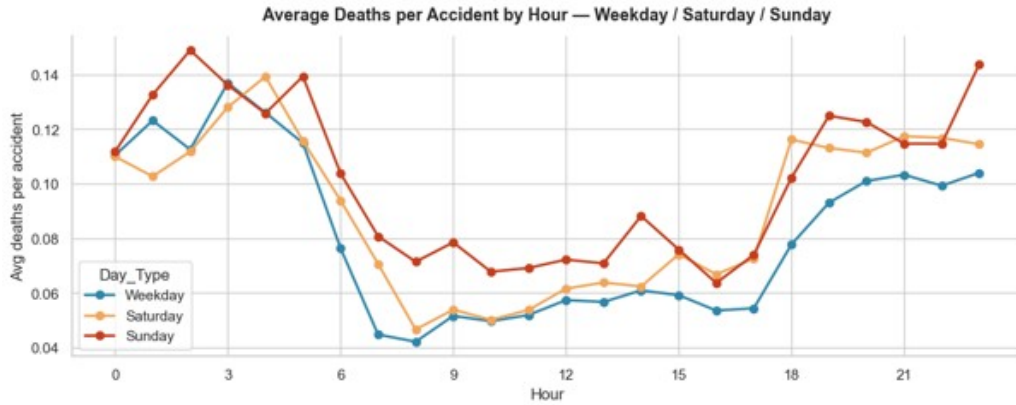


Figure 4: Hourly severity by Day_Type.

3.6.4 Accident Causes

Attribute Accident_Cause has over 80 categories. Most of them are similar and could be merged to create broader categories. For this purpose, 11 categories were created with the aim of developing a more robust and compact predictor. This suggests that half of all accidents are caused by drivers' behaviour on the road. However, it is interesting to see that these accidents are not the most frequent cause of death. The deadliest category is Pedestrian-related, which contains 30.9% of the total fatal share. Next is Driver state (fatigue/medical/distraction) and Visibility & signage, which over the overall fatal share as well. The surprising categories, which are below the index in Figure 5, are Alcohol/Drugs and Vehicle defect. Based on advertising and public opinion, it would be expected that these categories would rank higher or perhaps these safety measures are helpful, which is why these categories are lower than one would think. This does not contradict the findings of the study [Ahmed et al. \(2023\)](#), where alcohol and drugs were strong predictors for New Zealand crash data, because it secured third place in the table in terms of the number of accidents. This discovery also confirms the findings of the study [Scholz et al. \(2024\)](#), which states that accidents involving pedestrians have more serious consequences. Nevertheless, there is a high chance that Cause_Group will be an important predictor.

Table 8: Accident causes by volume, with fatal share for each category.

Cause_Group	Accidents	Share of all	Fatal share
Driver behaviour	227,662	49.2%	6.5%
Speed & following distance	75,854	16.4%	5.8%
Alcohol / Drugs	36,074	7.8%	5.4%
Vehicle defect	34,529	7.5%	2.9%
Road condition / infrastructure	28,601	6.2%	4.2%
Driver state (fatigue / medical / distraction)	21,614	4.7%	9.4%
Pedestrian-related	14,569	3.1%	30.9%
Animals	10,533	2.3%	5.0%
Visibility & signage	5,372	1.2%	8.4%
Weather	4,525	1.0%	4.7%
Other	3,800	0.8%	5.6%

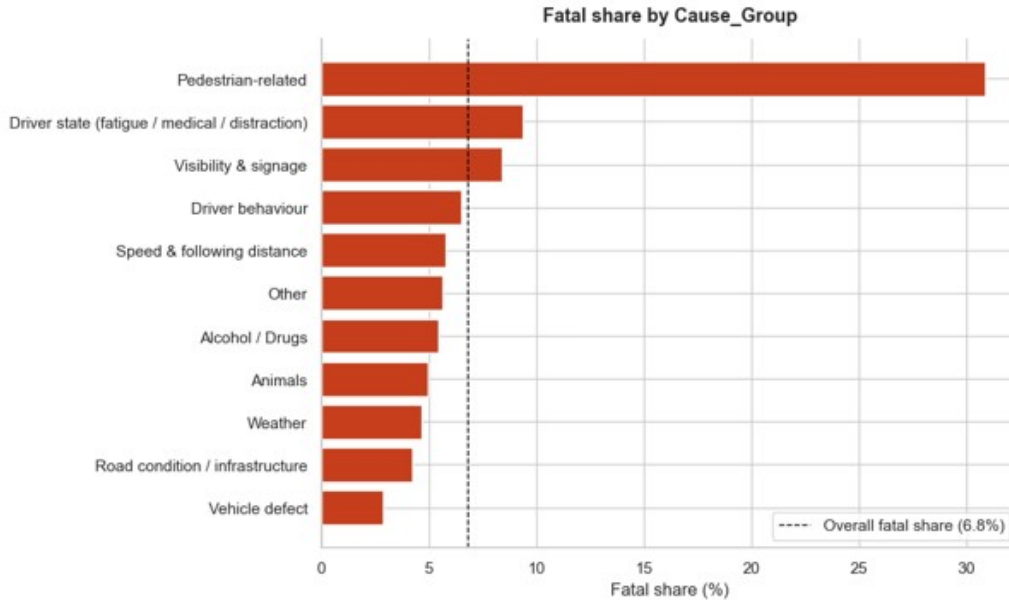


Figure 5: Fatal share by attribute Cause_Group.

3.7 Artefact Design

This is a design for the final outline of the process that will be implemented. This will be more detailed and explained in the Section 4.

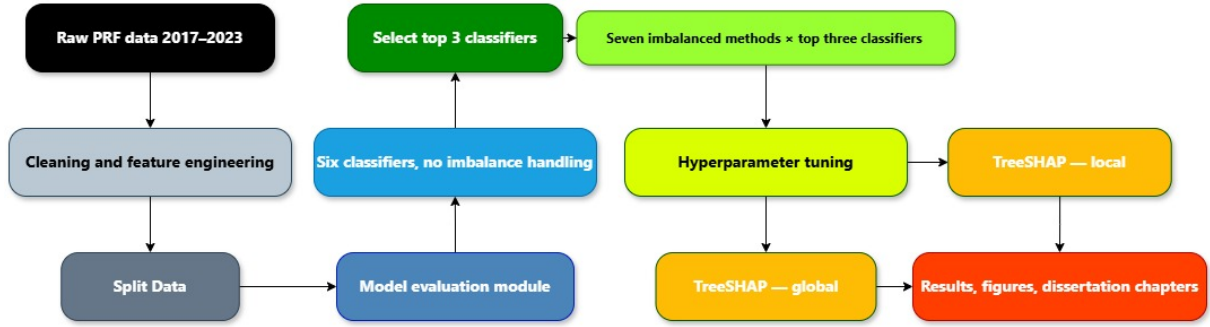


Figure 6: Overall artefact development pipeline

4 Proposed Future Process

This section is suggesting and explaining the next steps going to be taken during the summer of 2026. This work will need to go through six phases, each of which is connected in some way to the information obtained through literature review, EDA, or in relation to the research questions.

4.1 Phase 1 – Finalising the dataset (mid/late June)

This phase is going to focus on the cleaned dataset. It is going to decide to drop some unnecessary attributes, as mentioned before in Section 3, and save the clean dataset. The next step would be to split the data into training and test sets. Once done, the reusable evaluation module will be created, which contains all relevant metrics such as accuracy, macro-F1, per-class F1, MCC, and confusion matrix. This phase doesn't answer any research question in particular, but it lays a solid foundation for the subsequent sections.

4.2 Phase 2 – Base modelling (late June)

Based on Section 2, it was decided to use six classifiers. In this phase, no imbalance-handling technique is being applied. With the evaluation module from Phase 1, three classifiers are going to be selected based on their macro-F1 score. These top three classifiers are going to be used for the next Phase 3. The six classifiers to be compared are:

- **Logistic Regression**
- **Decision Tree**
- **Random Forest**
- **XGBoost**
- **LightGBM**
- **CatBoost**

This phase is linked to the research question 1. It should answer the question of whether non-linear machine learning models' predictions are better than those of a linear model.

4.3 Phase 3 – Imbalance-handling experiments (mid/late July)

The three best classifiers from phase 2 are going to be selected, and theoretically, seven imbalance-handling techniques are going to be applied on them to improve the performance. The seven techniques are:

- **none (basic model)**
- **random oversampling**
- **random undersampling**
- **SMOTE**
- **ADASYN**
- **SMOTE+Tomek**
- **class weighting**

This is the main contribution of the dissertation and a contributor to the answer to the research question 2.

4.4 Phase 4 – Hyperparameter tuning (early August)

In addition, the hyperparameter tuning process will be used to reach the proper conclusion. Only the top two or three combinations of methods and classifiers are going to be tuned. Python library Optuna will be used to search for the optimal parameter with stratified CV. The goal would be to find the best result for macro-F1. This should enhance the answer to the research question 2.

4.5 Phase 5 – SHAP analysis (early/mid August)

Based on the best performing model from phase 4, the TreeSHAP is going to be used for assessing the most important feature based for the global ranking and local explanation for each severity class. The final conclusions will be compared with previous studies, such as the results reported in [Ahmed et al. \(2023\)](#) and [Scholz et al. \(2024\)](#). This should lead to answering the final research question 3.

4.6 Phase 6 – Writing and submission (late August–early September)

In the last phase, the main objective will be to properly write down the findings and results. Utilise the chapters from this report and complete the additional chapters, such as the methodology, results, discussion, and conclusion. This is the final piece of work, which needs to be submitted in early September with a complete artefact and the presentation.

4.7 Final words

During the summer, three weeks break (from the end of June to the middle of July) is going to be taken, where no updates are going to be visible on GitHub. This should be mentioned to understand the gap in commits. Overall, the time planned for this work will vary depending on the situation. Therefore, the possibility of finishing the particular phase early or later than expected is most likely to happen.

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A Project Links

GitHub repository:	GitHub Link
DkIT Mahara:	Mahara Link

B Prompts for Generative AI

Prompt 1

Create a function that categorizes accident causes by keyword into fewer than 85 groups. The key is to group similar cases together to avoid duplication, and then specify the percentage of accidents that fall into each group. These groups are listed below.

Accident_Cause	
Driver's lack of attention to conveyance	107788
Incompatible velocity	43722
Driver's lack of reaction	40883
Driver broke the laws of transit	29318
Alcohol consumption	22140
Disrespect of safe distance from the next car	17929
Driver was sleeping	16820
Mechanical loss/defect of vehicle	15850
Acessing the road without seeing the presence of other vehicles	14973
Slippery track	12648
Driver failed to keep distance from the vehicle in front	12262
Alcohol ingestion by the driver	11229
Animals on the road	10533
The driver passed the next car improperly	9943
Lane change maneuver	9905
Pedestrian's lack of attention	8076
Excessive use of the car's tire	7566
Electrical or mechanical flaws	7246
Driver was in the opposite direction	5418
Road's defect	4487
Disrespecting the intersection	4106
Static object on the drainage gate	3326
Rain	3095
Cardiac attack	2951
Visibility restriction	2403
Excessive load/cargo	2364
Traffic with a motorcycle (or similar) between lanes	2290
Prohibited conversion	2266
Alcohol and/or drug ingestion by the pedestrian	2138
Pedestrian was walking in the road	1780
Unexpected pedestrian entry	1766
Irregular access	1705
Pedestrian was crossing the road outside of the crosswalk	1583
Accumulation of water on the road	1533
Driver had a cardiac attack	1515
Natural phenomena	1430
Deficiency of vehicle's sinalization/ilumination system	1258
Inadequate sinalization of the road	1193
Roads with holes without cement	1116
Abrupt use of the car's brake	1095
External fight	1084

Disobedience to laws of transit by the pedestrian	1055
Car's brake problem	846
Driving on the breakdown lane	827
Driver changed the lane illegally	778
Driver disrespected the red traffic light	741
Curvy road	726
Other flaws/problems in the road	660
Road had lots of sand/wreckage	651
Absence of sinalization	526
Sinking or ondulation in the pavement	483
Uneven breakdown lane	480
Oil accumulation on the road	437
Poor illumination (of the road)	411
Stopping at a prohibited place	403
Road works (in maintenance)	342
Drug ingestion	331
Driver using cellphone	322
Urban area without appropriate pedestrian walking	275
Obstacle in the road	238
Driver was using drugs	236
No breakdown lanes	234
Unlevel track	180
Fog	170
Car's suspension system with problems	163
Temporary return	121
Bad positioning of traffic sign posts	118
driving on the breakdown lane	114
Prominent downhill	90
Smoke	87
Visibility restrictions in horizontal curves	71
Pedestrian crosswalk with insufficient width	34
prohibited modification (of the car)	31
car's on sidewalk	31
Failed to activate the motorcycle headlight (or similar)	31
Visibility restrictions in vertical curves	30
speed reducer wasn't working correctly	25
engaging in drifting	24
Traffic lights with problems	22
unregulated headlights	20
hidden/covered sinalization (of the road)	11
robbery attempt	10
mental disorder (except suicide)	6

suicide (presumed)	5
inefficiency of the drainage system	4

Prompt 2

I got this Brazilian traffic accident data. Each row has a State column with those 2-letter codes.

State	
MT	1.230
BA	1.229
MG	1.219
GO	1.217
MS	1.215
AC	1.212
MA	1.203
AP	1.194
ES	1.194
RO	1.194
RS	1.183
TO	1.179
PR	1.159
PI	1.156
PA	1.155
AL	1.152
SC	1.152
AM	1.138
RJ	1.126
PE	1.113
CE	1.111
RR	1.110
RN	1.105
SE	1.104
PB	1.101
SP	1.083
DF	1.080

Therefore for my EDA I need to group them into the 5 big regions Brazil uses (north, northeast, center-west, southeast, south). Could you also double check that all 27 states are there and nothing got put in two regions by accident.

Prompt 3

(Asked AI to create tables for Overleaf based on the tables from EDA, and personally modified them to be more suitable for the interim report. Here is the example:)

Could you create a table for Overleaf based on the output I got from my code?

	accidents	fatal_share	accident_share
Cause_Group			
Driver behaviour	227662	0.065	0.492
Speed & following distance	75854	0.058	0.164
Alcohol / Drugs	36074	0.054	0.078
Vehicle defect	34529	0.029	0.075
Road condition / infrastructure	28601	0.042	0.062
Driver state (fatigue / medical / distraction)	21614	0.094	0.047
Pedestrian-related	14569	0.309	0.031
Animals	10533	0.050	0.023
Visibility & signage	5372	0.084	0.012
Weather	4525	0.047	0.010
Other	3800	0.056	0.008